RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Sunday, 30 August 2026, 8:20 PM
State	Finished
Completed on	Sunday, 30 August 2026, 8:21 PM
Time taken	40 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Divide and Conquer recursive function to find two elements with given sum
int findTwoSum(int arr[], int low, int high, int x, int *res1, int *res2) {
    // Base Case: If the search space is exhausted or invalid
    if (low >= high) {
        return 0; // No such pair exists
    }

    int currentSum = arr[low] + arr[high];

    // Check if the current boundaries sum up to x
    if (currentSum == x) {
        *res1 = arr[low];
        *res2 = arr[high];
        return 1; // Pair found
    }

    // Divide phase based on the sum comparison
    if (currentSum > x) {
        // If current sum is too large, look into the smaller subarray by reducing hig
        return findTwoSum(arr, low, high - 1, x, res1, res2);
    } else {
        // If current sum is too small, look into the larger subarray by increasing lo
        return findTwoSum(arr, low + 1, high, x, res1, res2);
    }
}

int main() {
    int n;

    // Read the size of the array
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 1;
    }

    int* arr = (int*)malloc(n * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    // Read the next n lines for array elements
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    // Read the target sum value x
    scanf("%d", &x);
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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